Home Network Security Assessment Report

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A comprehensive home network security assessment was conducted to evaluate the security posture of a residential network. The assessment focused on identifying network configuration settings, evaluating firewall protections, and implementing security hardening measures on a Verizon router.

The findings revealed that the network had moderate baseline security, which was improved through configuration changes including enabling high firewall protection and upgrading wireless encryption to WPA3.

Network Overview

Device IP Address: 192.168.1.230

Default Gateway(Router): 192.168.1.1

Router Type: Verizon Internet Gateway

Network Range: 192.168.1.0/24

Methodology

- Identified the local p address using command -”ipconfig getifaddr en0”
- Identified the default gateway using - “netstat -nr | grep default”
- Accessed router configuration panel via web interface
- Reviewed firewall, wireless, and system configuration settings
- Applied security hardening configurations

```
jbbpm@TTLT-MAC-13 ~ % ipconfig getifaddr en0
192.168.1.228
jbbpm@TTLT-MAC-13 ~ % netstat -nr | grep default
default          192.168.1.1       UGScg               en0
default          link#23           UCSIg          bridge100     !
default          link#25           UCSIg          bridge101     !
default                              fe80::aaa2:37ff:fe85:e242%en0         UGcg               en0
default                              fe80::%utun0                          UGcIg            utun0
default                              fe80::%utun1                          UGcIg            utun1
default                              fe80::%utun2                          UGcIg            utun2
default                              fe80::%utun3                          UGcIg            utun3
jbbpm@TTLT-MAC-13 ~ %
```

Findings

Finding 1: Initial wifi security was set to normal

Risk: Moderate exposure to inbound threats due to less restrictive filtering

Severity: Medium

Finding 2: WiFi encryption was reviewed and updated to WPA3

Risk: Older encryption protocols (WPA2 or below) are more vulnerable to attacks

Severity: Medium

Finding 3: Router required password authentication for access and login attempt limits were configured

Risk:if default credentials are used, unauthorized access is possible

Severity: Medium

Remediation Actions Taken

The following security improvements were successfully implemented:

Upgraded firewall from Normal to High Security

- Increased protection by blocking more inbound traffic

Enabled WPA3 wireless encryption

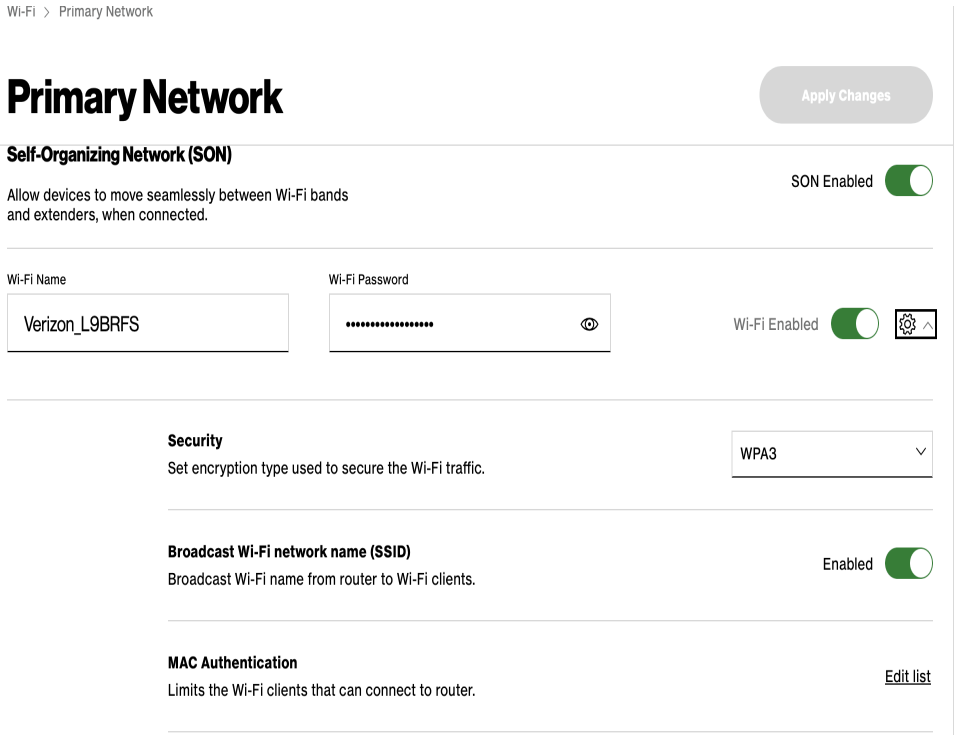
- Strengthened WiFi security against modern attacks

Verified password protection for router access

- Ensured authentication is required for configuration changes

Confirmed login attempt limits

- Helps prevent brute-force attacks



Security & Firewall > General

General

Apply Changes

IPv4 Settings

High Security
Remote administration will overide the security inbound policy

Inbound Rule | Outbound Rule

Outbound Set Top Box Traffic disabled

Normal Security
Remote administration will overide the security inbound policy

Inbound Rule | Outbound Rule

Low Security
Remote administration will overide the security inbound policy

Inbound Rule | Outbound Rule

IPv6 Settings

Network Devices

System > System Settings

Verizon Internet Gateway

System Settings

App

Diagnostics & Monitoring

System

System Status

Date & Time

Factory Reset

LED Brightness

Open Source Software

Reboot Router

Remote Administration

System Settings

User name Admin

Set new password •••••••••••••• minimum 8 chara

Retype new password ••••••••••••••

Unsuccessful Login Attempts 8 maximum attempts

Router

☐ **Automatic Refresh of System Monitoring Web Pages**

☑ **Prompt for Password When Accessing via LAN**

☑ **Warn User Before Configuration Changes**

Security Recommendations

To further improve network security:

- Change router admin password regularly and ensure it is strong
- Disable remote administration to prevent external access
- Keep router firmware updated to patch vulnerabilities
- Disable unnecessary services (e.g., UPnP, WPS if enabled)
- Monitor connected devices regularly for unauthorized access

Conclusion

The home network was found to have a moderate level of security, which was significantly improved through firewall hardening and upgrading wireless encryption. With continued monitoring and implementation of recommended controls, the network can maintain a strong security posture against common threats.